KARAMJEET SINGH

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PROFESSIONAL SUMMARY

Finance final-year graduate and FRM Part 1 qualified analyst with hands-on experience in quantitative risk modelling, portfolio analytics, and equity & FX market analysis. Proficient in Python and Excel-based risk frameworks including VaR, CVaR, Monte Carlo Simulation, and Markowitz Optimization. Seeking a Risk Analyst / Market Risk Analyst / Financial Analyst role at a leading financial institution.

CORE SKILLS & COMPETENCIES

Risk & Quantitative: Value at Risk (VaR), CVaR / Expected Shortfall, Monte Carlo Simulation, Markowitz MVO, Efficient Frontier, Portfolio Backtesting, Stress Testing, Sharpe Ratio Optimization

Market & Asset Classes: FX & Derivatives (G10 Currencies, XAU/USD, DXY), Fixed Income, Cross-Asset Correlation Analysis, Technical Analysis (RSI, MACD, Bollinger Bands)

Tools & Platforms: Python, Streamlit, Advanced Excel, Power BI, MS Office

AI & Development: Prompt Engineering, AI-augmented development — designed, built, and deployed full-stack financial web applications using LLM-driven workflows

Finance Credentials: FRM Part 1 (GARP, 2025), Equity Market Analyst (Finlatics), Investment Banking Analyst (Finlatics)

PROFESSIONAL EXPERIENCE

Finance Intern – Trading & Market Analysis | **BB Advisory**

- Actively traded a portfolio of **14+ currency pairs** using systematic **backtesting and strategy optimisation**; achieved an **8% portfolio return** over the 3-month internship period.
- Conducted **cross-asset correlation analysis** on **DXY, 7 major G10 currencies, and Gold (XAU/USD)**; produced written trade rationale with entry/exit signals, risk-reward ratios, and stop-loss levels.
- Monitored **US equity indices (S&P 500, Nasdaq, Dow Jones)** and global benchmarks daily; applied **position sizing and volatility-adjusted exposure** to manage downside risk.

Finance Intern – Technical Research | **Bridge Wealth Solution**

- Performed **technical analysis** on **3 NSE-listed equities** using **RSI, MACD, and moving averages**; generated research notes with price targets and risk flags shared with senior analysts.

QUANTITATIVE PROJECTS

Portfolio Optimizer | *Markowitz MVO · Efficient Frontier · Live Deployment* *2025*

- Built an automated optimization engine applying the **Markowitz Mean–Variance framework** across a user-configurable multi-asset universe (equities, bonds, commodities) — autonomously fetches **10 years of live price data**, computes annualised return vectors and covariance matrices, and solves **Min Variance, Max Sharpe, and Max Return portfolios** via SLSQP constrained optimization; supports adjustable risk-free rate, and multiple return frequencies.
- On a sample AAPL/MSFT/GOOG portfolio, optimization improved the **Sharpe ratio from 1.02 → 1.84 (+80.4%)**, boosting expected return by **+32.28%** (27.88% → 60.16%) while containing volatility increase to **+7.12%**; benchmarks the **tangency portfolio vs. equal-weight strategy** with live Sharpe, volatility, and VaR (95%) comparison; deployed at **karamfrm.com.**

Tail Risk Analyzer | *VaR · CVaR · Monte Carlo · Live Deployment* *2025*

- Engineered a **multi-method VaR engine** supporting **Historical, Parametric, and Monte Carlo simulation (10,000 scenarios)** with configurable **confidence levels (90–99%) and horizons (1D–20D)** — fully automated pipeline: raw price data → log returns → covariance estimation → VaR/ES computation; on a sample portfolio, generated **5-Day VaR (95%) of $2,377 and ES of $3,461** (ES/VaR ratio: 1.46×).
- Implemented a **Risk Sensitivity Matrix** (3 confidence levels × 4 horizons × 3 lookback regimes), **tail amplification analysis (3.58×)**, and **regime sensitivity scoring (11.9%)**; deployed as a live web app at **var.karamfrm.com** with automated diagnostic insights and multi-portfolio comparison.

In-Progress: FIFA WC 2026 ML Prediction Model | Cross-Asset Correlation Heatmap (FX, Indices, Rates, Commodities)

RESEARCH PROJECT

Six Sigma Efficacy | *DMAIC · DMADV · Multi-Industry Analysis* *2024*

- Conducted in-depth research on the **efficacy, constraints, and business impact** of Six Sigma methodologies across **3 industries (Automotive, E-Commerce, Mining)**.
- Undertook **quantitative and qualitative analysis** of **DMADV and DMAIC frameworks**; decoded implementation outcomes and performance improvements across 3+ companies.

EDUCATION & CERTIFICATIONS

Financial Risk Manager (FRM) – Part 1 | GARP *Passed 2025*

BMS (Finance) | Shaheed Sukhdev College of Business Studies, University of Delhi *2022 – 2025 | CGPA: 7.77 / 10*

AISSCE (Commerce – Class XII) | S.S. Mota Singh School, New Delhi *2022 | 90.6% [453/500]*

Additional Certifications: Equity Market Analyst (Finlatics) | Investment Banking Analyst (Finlatics) | Advanced Excel (Udemy) | Digital Marketing (Google)